

Coseismic and Postseismic Deformation Associated with the 2007 Mw8.1 Solomon Islands Earthquake: A Joint InSAR and Coral Geodesy Analysis

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Key Points:

- This study developed a new InSAR analysis method to distinguish deformation signals from tropospheric noise.
- ALOS InSAR observations capture up to 235.8 cm of coseismic LOS deformation and a postseismic deformation rate decay of 26.4 to 5.9 cm/year over 2.5 years.
- InSAR results are consistent with vertical deformation time series inferred from field coral $\delta^{13}C$ measurements.

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Abstract

This study utilizes space borne Interferometric Synthetic Aperture Radar (InSAR) observations to investigate coseismic and postseismic surface deformation resulting from the 2007 Mw 8.1 Solomon Islands earthquake. This research has previously been limited by data coverage gaps and tropospheric noise contamination of the signal as a result of the region's humid climate and dense vegetation; however, this study's findings include an advanced method to detect signal from noise in regions with a small signal-to-noise ratio and limited external validation data. A coseismic LOS deformation map was created utilizing a phase-filling processing technique. The map captured a maximum coseismic deformation of 235.8 cm on Vonavona island near the epicenter. The proposed signal detection method enabled the identification of the true postseismic deformation signal over Vonavona and the generation of a time series of postseismic deformation rates using only nine postseismic SAR acquisitions. The postseismic time series over Vonavona followed an expected exponential decay, with deformation rates decaying from 26.4 cm/year to 5.9 cm/year over approximately 2.5 years. InSAR data are complemented by vertical measurements derived from coral $\delta^{13}C$ that span the entire earthquake cycle. Coral signal processing revealed an interseismic subsidence rate of approximately 4 cm/year, a coseismic vertical displacement of 66 cm, and a clear exponential decay of postseismic vertical measurements, dropping approximately 50 cm in 2.5 years. The complementary use of in situ coral and remote sensing data sets expanded the temporal horizon of this study to cover all three periods of the earthquake cycle.

Plain Language Summary

Utilizing spaceborne interferometric synthetic aperture radar (InSAR), this study maps coseismic surface deformation and generates a postseismic deformation rate time series from the 2007 Mw8.1 Solomon Islands earthquake. The postseismic data are contaminated by tropospheric error, thus a new technique to decipher signal from noise in low signal-to-noise ratio cases is proposed. This technique generated a postseismic time series that recorded an exponential decay of deformation rates. The coseismic deformation map recorded a maximum line-of-sight deformation, measured in the direction from the satellite toward the ground, of 235.8 cm. Only the last two of three periods in the earthquake cycle are captured with InSAR data. Complementary coral $\delta^{13}C$ measurements span all three periods of the cycle, allowing this study to further investigate vertical deformation trends in the decade preceding the earthquake. The $\delta^{13}C$ -derived time series indicates an annual subsidence rate of ~ 4 cm/year pre-earthquake and an exponential decay of vertical deformation post-earthquake. Implications of this study include (1) a new strategy to detect usable signal in areas characterized by large tropospheric error, (2) a complementary method through $\delta^{13}C$ analysis to study entire earthquake cycles, and (3) a greater understanding of the tectonic processes that occurred during the 2007 earthquake.

1 Introduction

Understanding the earthquake cycle is crucial for investigating the potential of future hazards (Scholz, 2002; Savage & Thatcher, 1992). The interseismic period is characterized by strain accumulation due to locking of the faults, which accounts for the largest portion of the cycle and often lasts tens to hundreds of years. The coseismic period occurs as the accumulated strain exceeds the friction which had maintained the locking, and energy is released in the form of seismic waves in a few seconds or minutes. The postseismic period is defined by continued deformation stemming from both afterslip in the near field and various forms of relaxation in the far field over a few days to several months or years. Subduction zones are characterized by complex fault geometry and frequent production of large-magnitude earthquakes, and thus can serve as ideal natural labora-

65 tories for investigating slip distributions and earthquake recurrence intervals. Therefore,
 66 quantifying surface deformation over the Solomon Islands, which is located in one of the
 67 world’s most active subduction zones, can contribute to improved observations about the
 68 earthquake cycle in the region and lead to additional constraints on the local fault ge-
 69 ometry.

70 Spaceborne Interferometric Synthetic Aperture Radar (InSAR) has been used to
 71 reconstruct millimeter to centimeter— surface deformations with tens to hundreds of me-
 72 ters of spatial resolution over broad areas (Massonnet et al., 1993). InSAR has been suc-
 73 cessfully employed to study the earthquake cycle, including research of interseismic strain
 74 accumulation, coseismic surface deformation, postseismic processes, and constraints of
 75 fault geometry for locations around the globe. For example, the North Anatolian fault
 76 in Turkey has been extensively surveyed by InSAR, including multiple studies focused
 77 on identifying interseismic rates, which are historically smaller and often more difficult
 78 to capture than those generated in the co- and postseismic periods (Wright et al., 2001;
 79 R. J. Walters et al., 2014; Delouis et al., 2002; Cakir et al., 2014; Kaneko et al., 2013).
 80 Both R. J. Walters et al. (2014) and Cakir et al. (2014) were able to identify interseis-
 81 mic slip rates with an uncertainty of only 3 mm/year utilizing InSAR. This high-accuracy
 82 InSAR data, combined with GPS measurements, enabled the generation of high-resolution
 83 velocity and strain rate fields, and the constraint of the interseismic locking depth. Mul-
 84 tiple groups also have utilized InSAR to monitor stress changes along the San Andreas
 85 and San Jacinto faults to understand potential earthquake hazards in California (E. Lind-
 86 sey et al., 2013; Lyons & Sandwell, 2003; Tong et al., 2013). Well-known for its tectonic
 87 activity, California has been the site of many InSAR studies for all portions of the cy-
 88 cle, many of which surround specific coseismic events, including but not limited to the
 89 1992 Hector Mine, 2004 Parkfield, 2014 South Napa, and 2019 Ridgecrest earthquakes
 90 (Fialko et al., 2001; Johanson et al., 2005; Wei et al., 2015; Guns et al., 2022). For co-
 91 seismic studies, the InSAR data is often used to generate surface displacement maps or
 92 inverted to generate a slip model to better constrain sub-surface geometry. Similarly, In-
 93 SAR has been employed to better understand tectonic processes at the Altyn Tagh fault
 94 near Tibet (Liu et al., 2018; Elliott et al., 2008; Bell et al., 2011; Wang & Wright, 2012;
 95 Wright et al., 2004; Ryder et al., 2007; Sijia et al., 2020). Ryder et al. (2007) and Sijia
 96 et al. (2020) investigated the postseismic period following a Mw 7.6 and Mw 6.5 earth-
 97 quake in the Tibetan Plateau, respectively, and captured the temporal evolution of the
 98 postseismic deformation. The studies additionally inverted the time series to analyze the
 99 relationship between the co- and postseismic slip, enabling them to determine the equiv-
 100 alent postseismic moment release, reported as a percentage of the coseismic moment. Be-
 101 yond directly studying the deformation during the three periods of the earthquake cy-
 102 cle, InSAR analyses can provide insight about future earthquake risks. R. Walters et al.
 103 (2009) employed InSAR and body-wave seismology to determine source parameters for
 104 the 2009 Mw 6.9 L’Aquila earthquake. These parameters were utilized to determine the
 105 resulting stress changes on nearby faults that did not rupture, and their findings indi-
 106 cated that several were brought closer to failure. E. O. Lindsey et al. (2015) investigated
 107 the 2015 Mw 7.8 Gorkha, Nepal earthquake and its Mw 7.3 aftershock utilizing ALOS-
 108 2/PALSAR-2 observations. Their investigation revealed the aftershock both extended
 109 the rupture area and left a 20 km gap where the fault did not slip. They estimated that
 110 this unslipped fault patch has the potential to generate a Mw 6.9 event. Such findings
 111 can provide key information for future hazard planning.

112 Despite the advantages of InSAR analysis for tectonic studies and its proven suc-
 113 cess elsewhere, the Solomon Islands remain poorly surveyed by InSAR, with only one
 114 published study that employed InSAR in the region. Miyagi et al. (2009) reported over
 115 2 meters of co-seismic deformation from the 2007 Mw 8.1 Solomon Islands earthquake.
 116 Due to the difficulty in unwrapping interferograms over island regions, each of the three
 117 ALOS radar paths was processed individually. They solved for the slip distribution of
 118 this earthquake based on InSAR co-seismic deformation estimates and limited field data,

which suggested that the largest slip was around the hypocenter and centroid. Based on their slip model, they concluded that (1) the largest slip may be related to the subduction zone between the Woodlark and Australian plates, and (2) the smaller magnitude slip between the hypocenter and centroid may be attributed to thermal activity from the volcano located on Simbo Island. While many others have studied the tectonics of the region, few have employed InSAR in their investigations. The lack of prior InSAR studies over the Solomon Islands is attributed to the humid climate and dense vegetation of the region, which often leads to severe tropospheric errors and phase decorrelation noise in InSAR phase measurements. Furthermore, the availability of in situ measurements is extremely limited, which makes validating noisy InSAR results difficult. In this study, it is shown that L-band InSAR data can maintain phase coherence over 1.75 years over the densely vegetated Solomon Islands. This enables the detection of both coseismic and postseismic deformation of the 2007 Solomon Islands earthquake. Due to the lack of in-situ data, a new InSAR analysis method is developed to identify InSAR deformation signals from tropospheric noise based on their distinct temporal and spatial characteristics. Furthermore, this study presents vertical deformation time series derived from $\delta^{13}C$ measurements of coral skeleton samples, which offers a consistent and complementary field data set that expand the temporal horizon of previous 2007 Solomon Islands earthquake studies back to 1997. This enables measurement of the interseismic deformation prior to the 2007 Solomon Islands earthquake based on coral samples for the first time.

2 Background

The Solomon Islands reside in an active plate boundary zone between the Australian and Pacific plates. Based on GPS campaigns, significant rates of oblique convergence (~ 10 - 11 cm/year) occur between these two plates (Demets et al., 1990; Wallace et al., 2004; M.-C. Chen et al., 2014). Additionally, the region sees counterclockwise rotation of the Woodlark plate away from the Australian plate ($2.4^\circ/\text{Ma}$ for 0.5 Ma to present) (Weissel et al., 1982; Benes et al., 1994; Wallace et al., 2004, 2014). The main boundary between the Pacific and Woodlark plates is the Woodlark basin, which is an active extensional basin where a new oceanic crust is forming and spreading the sea floor. This third form of motion in the region is characterized by spreading rates ranging from 4 - 5.6 cm/year in the eastern Woodlark spreading center (Wallace et al., 2014). Spanning the island chain of interest (Figure 1), the New Britain-San Cristobal trench is classified as the most seismically active subduction zone in the Pacific. The San Cristobal trench, just west of the islands of interest, experiences convergence rates of ~ 9 - 10 cm/year (Wallace et al., 2014) and the New Britain trench, northwest of the islands of interest, experiences convergence rates of ~ 6 - 16 cm/year (Wallace et al., 2004). This coupling of large-scale motion across several plates and trenches distinguishes the Solomon Islands as an ideal natural laboratory for studying earthquake cycles.

Given the tropical nature of the region, the presence of coral microatolls on ring islands in the Solomon Islands provides a method to reconstruct the vertical deformation history that spans tens to hundreds of years into the past. Coral species in the region grow both upward and horizontally; however, upward growth is limited by water level (exposure to air and direct sunlight). The portion of the coral that emerges above the ocean surface dies, while the portion remaining below the surface can continue to grow horizontally. Thus, subsidence of the ocean floor increases the maximum possible vertical growth of the coral and uplift can result in notable stepped die-downs if the vertical deformation is large enough to result in the coral emerging above the ocean surface. The effects of subtle deformation require long time periods to be measured as the coral grows at ~ 1 cm/year, while rapid uplift and subsidence during a major earthquake can be captured with only a few days (Taylor et al., 1987). For example, Taylor et al. (2008) utilized coral microatoll to investigate the April 2007 catastrophic Mw 8.1 earthquake that ruptured just west of the New Georgia island chain. Their study analyzed

the vertical coral growth and die-downs to measure the associated coastal coseismic uplift and subsidence—depending on the location of the sample relative to the trench—a phenomenon that is rarely seen outside of this unique tectonic setting. Their coseismic deformation measurements suggest that the 2007 earthquake broke through a triple junction where the Australian and Woodlark plates subduct beneath the overriding Pacific plate. Coral microatoll have been used similarly to study coseismic vertical deformation in other regions that have experienced large magnitude earthquakes, including Haiti and Sumatra (Weil-Accardo et al., 2016; Meltzner et al., 2010; Zachariasen et al., 1999; Briggs et al., 2006).

3 Method

3.1 InSAR Data Processing

ALOS PALSAR scenes acquired between December 28, 2006 and January 22, 2010 over the area of interest (Figure 2a) were downloaded from the Alaska Satellite Facility (Table 1). All SAR scenes were processed using a motion-compensation imaging radar processor (Zebker et al., 2010), and no spatial or temporal baseline constraints were applied during interferogram formation. All interferograms were multi-looked to 240-meter pixel spacing, and the topographic phase component was removed using a Digital Elevation Model (DEM; 30 m pixel-spacing) derived from the Shuttle Radar Topography Mission (SRTM) (Farr et al., 2007). Interferograms acquired on the same day from the same path were merged by minimizing the phase difference between these overlapping regions of the adjacent frames, and the Goldstein filter was applied to merged interferograms to improve spatial phase coherence (Goldstein & Werner, 1998).

The interferograms were unwrapped using Statistical-cost, Network-flow Algorithm for Phase Unwrapping (SNAPHU) algorithm (C. W. Chen & Zebker, 2001). SNAPHU attempts to unwrap InSAR phase measurements at all pixels including bodies of water where phase is completely decorrelated and random in space. This can dramatically increase computing time and introduce unwrapping errors at nearby land pixels. To overcome this challenge, land and ocean pixels were classified using an average phase coherence threshold of 0.11, and then wrapped phase values at ocean pixels were replaced by wrapped phase values at nearby land-pixels (e.g., Shanker and Zebker (2009)). As shown in Figure 3, the wrapped phase value ϕ_{ocean} at an ocean pixel of interest was replaced by the weighted sum of wrapped phase values at nearby land pixels as:

$$\phi_{ocean} = \frac{\sum w_i \phi_i}{\sum w_i} \quad (1)$$

where ϕ_i is the wrapped phase of the i^{th} land pixel within a square box of length 11 pixels centered at the ocean pixel, $w_i = 1/r_i$, and r_i is the distance between the ocean pixel of interest and the i^{th} land pixel within the square box. This phase filling process was performed for a second time to further extrapolate phase observations at land pixels into ocean area, which reduces large phase discontinuities at the coastline and phase unwrapping errors at land pixels. After unwrapping, the land-ocean mask was employed to identify land pixels for surface deformation analysis.

At a land pixel of interest, InSAR unwrapped phase ϕ measures surface deformation Δd_{LOS} along the radar line-of-sight direction with respect to a reference pixel as:

$$\phi \approx \frac{4\pi}{\lambda} \Delta d_{LOS} = \frac{4\pi}{\lambda} (e_1 \Delta d_e + e_2 \Delta d_n + e_3 \Delta d_u) \quad (2)$$

where λ is the radar wavelength. Δd_e , Δd_n , and Δd_u are surface deformation in east, north and vertical directions, and $\mathbf{e} = [e_1, e_2, e_3]$ is the LOS unit vector. For the ALOS PALSAR data over the Solomon Islands, $\mathbf{e} = [0.6, 0.13, -0.79]$ at the mid-swath, and the look angle variation is not substantial across the ALOS PALSAR scenes. coseismic

deformation measurements (in east, north and vertical directions) of the April 1, 2007 Solomon Islands earthquake were recorded at three campaign GPS stations RAVA, MUND, and HUSU (Figure 2a). The GPS station RAVA (8.721°S 157.407°E) was chosen as the reference point to calibrate the InSAR coseismic deformation map. The other two stations were used to validate InSAR results. postseismic GPS time series were recorded at MUND for a brief period from April 23rd, 2007 through May 5, 2007. Given that postseismic deformation rate changes over time, postseismic interferograms were not calibrated to absolute deformation rates due to the lack of GPS data over the entire InSAR study period. Instead, postseismic interferograms were used to calculate the relative LOS deformation that occurred after the earthquake with respect to a reference point (8.202°S 157.355°E) on New Georgia island away from the epicenter and the coast.

3.2 Coseismic and Postseismic LOS deformation solutions

As shown in Equation (2), InSAR calculates the phase difference between two SAR acquisitions, from which the LOS surface deformation between these two acquisition dates can be derived (Rosen et al., 2000). Given that earthquakes occur over a short period of seconds to minutes and the ALOS satellite revisit time is 46 days (the acquisitions were irregular with data gaps), the coseismic deformation signal can be modeled as a step function (Figure 4a). In this study, the coseismic deformation signal was captured by a coseismic interferogram formed using one SAR acquisition immediately preceding and one SAR acquisition immediately following the earthquake that occurred on April 2, 2007. For path 344, the coseismic interferogram spans March 1, 2007 and April 16, 2007. For path 343, the coseismic interferogram spans February 12, 2007 and May 15, 2007. The magnitude of the expected coseismic deformation signal is on the order of meters (Miyagi et al., 2009; Taylor et al., 2008), which is well above the InSAR noise level. As a result, averaging multiple coseismic interferograms that span the 2007 Solomon Islands earthquake is not necessary.

The postseismic signal is expected to follow an exponential or logarithmic decay (Figure 4b) depending on the postseismic mechanism at play (Sobrero et al., 2020; Yusiyaniti et al., 2023). Because the magnitude of the postseismic deformation is significantly smaller than the coseismic deformation, a stacking approach (Sandwell & Price, 1998; Staniewicz et al., 2020) was employed to calculate the average LOS velocity, $\bar{v}_k$, for the k^{th} period of interest after the 2007 Solomon Islands earthquake:

$$S_k = \frac{\lambda}{4\pi} \frac{\sum_{i \in G_k} \phi_i}{\sum_{i \in G_k} \Delta t_i} = \frac{\sum_{i \in G_k} \Delta d_{LOS,i}}{\sum_{i \in G_k} \Delta t_i} + \delta = \bar{v}_k + \delta \quad (3)$$

where S_k denotes the k^{th} stack, $\lambda = 24$ cm is the ALOS PALSAR radar wavelength, and G_k is a set of interferograms that were formed using two SAR scenes acquired within the k^{th} period of interest. ϕ_i , $\Delta d_{LOS,i}$, and Δt_i are the unwrapped phase, the corresponding LOS deformation, and the temporal baseline of the i^{th} interferogram in G_k , respectively. δ is the residual noise term.

Nine ALOS PALSAR path 344 scenes acquired between April 16, 2007 and January 19, 2010 were used in the postseismic deformation analysis (Figure 4b). Here the postseismic deformation analysis is limited to every land pixel in path 344 as this path is closer to the 2007 earthquake epicenter than path 343, and the expected signal-to-noise ratio (SNR) is better. Following Equation (3), the average LOS velocity was calculated over three consecutive periods of interest: (1) April 16, 2007 through October 17, 2007; (2) October 17, 2007 through April 18, 2008; and (3) April 18, 2008 through January 22, 2010. The interferograms used in each consecutive stack are listed in Table 2. Furthermore, the average LOS velocity was calculated over the following growing time windows starting on April 16, 2007 through: (1) September 1, 2007 (2 acquisitions and 1 interferogram); (2) October 17, 2007 (3 acquisitions and 3 interferograms); (3) January 17, 2008 (4 acquisitions and 6 interferograms); (4) March 3, 2008 (5 acquisitions and 10

interferograms); (5) April 18, 2008 (6 acquisitions and 15 interferograms); (6) September 3, 2008 (7 acquisitions and 21 interferograms); (7) January 19, 2009 (8 acquisitions and 28 interferograms); and (8) January 22, 2010 (9 acquisitions and 36 interferograms). Section 3.3 further presents a method to determine whether postseismic deformation signals derived from these stacks of interferograms are above the residual noise level and thus detectable.

3.3 InSAR Measurement Uncertainty Quantification

The unwrapped phase at a pixel of interest, ϕ , can be written as:

$$\phi = \frac{4\pi}{\lambda} \Delta d_{LOS} + \Delta\phi_{DEM} + \Delta\phi_{decorr} + \Delta\phi_{unwrap} + \Delta\phi_{orb} + \Delta\phi_{iono} + \Delta\phi_{tropo} + \Delta\phi_n \quad (4)$$

where λ is the radar wavelength, and Δd_{LOS} is the LOS deformation between the two SAR acquisitions used to form the interferogram. The remaining terms are the result of various error sources: inaccurate DEM ($\Delta\phi_{dem}$), phase decorrelation ($\Delta\phi_{decorr}$) and the resulting unwrapping errors ($\Delta\phi_{unwrap}$), orbital errors ($\Delta\phi_{orb}$), ionospheric ($\Delta\phi_{iono}$) and tropospheric ($\Delta\phi_{tropo}$) artifacts, and other error terms ($\Delta\phi_n$) such as changes in soil moisture.

In this study, the magnitude of the observed InSAR phase does not correlate with InSAR perpendicular baseline, a key signature of DEM errors (Fattahi & Amelung, 2013). As a result, the impact of DEM errors is minimal. Based on visual inspection, all interferograms formed using path 344 scenes acquired on two dates (January 14, 2007 and October 19, 2008) suffer from severe phase decorrelation artifacts and relatively large unwrapping errors. This is likely due to changes in surface scattering properties on these two days (e.g., due to a severe weather event). These two SAR acquisitions and the associated interferograms were excluded in this study, and a total of nine high-quality path 344 SAR acquisitions were used in the postseismic deformation analysis (Figure 4b). For all interferograms formed using these nine path 344 acquisitions, InSAR phase coherence is well maintained regardless of temporal baselines, except over Tetepare Island, which lies southeast of Rendova. Consequently, the postseismic deformation analysis focuses on Vonavona, Kohinggo, Kolombangara, and New Georgia, where decorrelation artifacts $\Delta\phi_{decorr}$ and unwrapping errors ($\Delta\phi_{unwrap}$) are not substantial. A planar phase ramp was estimated and removed in each postseismic interferogram to account for orbital errors ($\Delta\phi_{orb}$) and long-wavelength tropospheric and ionospheric noise. Given that the orientation of the observed phase ramp in each interferogram varies, these ramps are unlikely related to true deformation signals. The other error sources, $\Delta\phi_n$, associated with thermal and soil moisture effects, are on the order of millimeters and much smaller than other noise terms (Michaelides et al., 2019). Thus, it is concluded that the dominant error term is tropospheric noise in the Solomon Islands ALOS PALSAR data set.

Given that tropospheric error is the dominant noise term in the signal over the Solomon Islands region, Equation (4) can be simplified to:

$$\phi_{ij} = \frac{4\pi}{\lambda} (\Delta d_{ij} + n_j - n_i) \quad (5)$$

where λ is the radar wavelength and Δd_{ij} is the LOS surface deformation between two SAR acquisitions acquired on dates i and j . The tropospheric noise in the LOS direction on each acquisition date is represented as n_i and n_j , respectively.

In this study, nine ALOS PALSAR path 344 scenes were used in the postseismic deformation analysis (Figure 4b). First, consider the stacking results over three consecutive periods of interest: (1) April 16, 2007 through October 17, 2007; (2) October 17, 2007 through April 18, 2008; and (3) April 18, 2008 through January 22, 2010. Following Equation (3) and Equation (5), each of the three stacks defined in Table 2 can be expressed as:

$$S_1 = \bar{v}_1 + \frac{2n_3 - 2n_1}{\sum_{i=1, i < j}^3 \Delta t_{ij}} \quad (6)$$

$$S_2 = \bar{v}_2 + \frac{3n_6 + n_5 - n_4 - 3n_3}{\sum_{i=3, i < j}^6 \Delta t_{ij}} \quad (7)$$

$$S_3 = \bar{v}_3 + \frac{3n_9 + n_8 - n_7 - 3n_6}{\sum_{i=6, i < j}^9 \Delta t_{ij}} \quad (8)$$

Here $\bar{v}_1$, $\bar{v}_2$, and $\bar{v}_3$ are the average deformation rates for each of the three periods of interest. n_i is the LOS tropospheric noise of the i^{th} SAR acquisition ($i = 1, \dots, 9$), and the total time span of all interferograms in the three stacks are 1.01, 1.64, and 5.67 years, respectively. In the case where postseismic deformation signals are well above the noise level, a consistent deformation pattern should be observed in all three stacking solutions. At a given location, the observed deformation rate is expected to decrease in time (e.g. a larger rate is expected in S_1 than S_3). For a given stack solution S_k , the observed deformation rate is expected to decrease as the distance to the epicenter increases. In contrast, tropospheric errors are not correlated for time increments greater than one day (Zebker et al., 1997; Emdarson et al., 2003). Given that the ALOS PALSAR satellite revisit cycle is 46 days, $n_1, \dots, n_9$ are temporally uncorrelated. This suggests that the residual tropospheric noise patterns in S_1 and S_3 are not correlated. The residual noise level in S_3 tends to be smaller than in S_1 because the total time span of all interferograms in S_3 (5.67 years) is more than five times larger than in S_1 (1.01 years). Furthermore, the third SAR acquisition (October 17, 2007) was used to form interferograms in both S_1 and S_2 . As shown in Equation (6) and Equation (8), the contribution of n_3 in S_1 and S_2 has the opposite sign. Similarly, the contribution of n_6 in S_2 and S_3 also has the opposite sign. In summary, the temporal and spatial characteristics of tropospheric noise and postseismic deformation signals are different. This allows determination of whether postseismic deformation signals are detectable by a comparison of S_1 , S_2 , and S_3 .

Next, consider the following growing time windows starting on April 16, 2007 through: (1) September 1, 2007 (2 acquisitions); (2) October 17, 2007 (3 acquisitions); (3) January 17, 2008 (4 acquisitions); (4) March 3, 2008 (5 acquisitions); (5) April 18, 2008 (6 acquisitions); (6) September 3, 2008 (7 acquisitions); (7) January 19, 2009 (8 acquisitions); and (8) January 22, 2010 (9 acquisitions). For the case of M SAR acquisitions that all $N = M \times (M - 1)/2$ interferograms are formed, the stacking result, S_M , can be written as:

$$S_M = \frac{1}{\sum_i \Delta t_i} \left(\sum_i^N \Delta d_{LOS,i} + \sum_m^M w_m n_m \right) \quad (9)$$

where $\Delta d_{LOS,i}$ and δt_i are the LOS deformation and the temporal baseline of the i^{th} interferogram, respectively. n_m is the tropospheric noise of the m^{th} SAR acquisition, and the weight applied to the tropospheric noise of the m^{th} SAR scene, w_m , can be described as

$$w_m = 2m - (M + 1) \quad (10)$$

Equation (9) and Equation (10) quantitatively describe the stacking results of eight growing time windows, where each stack contains an additional SAR scene forward in time. For $M = 2$, $w_1 = -1$ and $w_2 = 1$. For $M = 9$, the largest absolute weights are $w_1 = -8$ and $w_9 = 8$. Furthermore, as M increases, the total time span of all interferograms increases. For the Solomon Islands case, the total time space for $M = 2$ is 0.378 years,

while the total time space for $M = 9$ is 34.5 years. This means that the residual tropospheric noise from the 1st SAR acquisition (April 16, 2007) is more than 11 times larger when $M = 2$ than when $M = 9$. In this stacking design with growing time windows, both the postseismic deformation signal and n_1 are expected to produce a spatially consistent pattern with reduced magnitude over time. In contrast, residual tropospheric noise from the remaining SAR acquisitions is expected to produce random patterns in these eight stacking results. To further separate the contribution from the postseismic deformation signal and n_1 , an alternative stacking test excluding all interferograms formed using the 1st SAR acquisition (April 16, 2007) can be calculated, where n_1 does not contribute to the result.

3.4 $\delta^{13}\text{C}$ Time Series

The $^{13}\text{C}/^{12}\text{C}$ ratio ($\delta^{13}\text{C}$) signal preserved in shallow-living coral microatolls provides a valuable archive of past sea-level fluctuations. Analysis of X-rayed cross-sections of coral heads (Figure 5) reveals alternating positive and negative changes in relative sea level. These fluctuations are expressed as die-down surfaces and subsequent upgrowths at the highest level of survival (HLS). During sustained emergence, exposed coral surfaces die, and growth continues laterally and upward at the HLS. When relative sea level rises, coral colonies grow vertically toward the sea surface at rates ranging from 1 to 1.5 cm/yr.

A striking example is provided by the M_W 8.1 earthquake of April 1, 2007, which produced dramatic vertical coastal displacements. Field surveys of coral reefs documented 0.73 ± 0.14 m of coseismic subsidence at Simbo Island, and up to 2.46 ± 0.14 m of coseismic uplift at Ranongga Island, located only 8 km away on the overriding plate (Taylor et al., 2008). Annual density growth bands in corals, analogous to tree rings, allow band counting and provide an internal chronometer with annual resolution. This technique enables dating of events with a precision of approximately ± 1 year relative to the living surface at the time of collection (Buddemeier & Taylor, 2000). Such precision makes it possible to reconstruct the timing and magnitude of both uplift and subsidence for living or fossil corals (Biemiller et al., 2020).

The cross-section used in this study (Figure 5) was retrieved in April 2012 from a *Porites lutea* microatoll that had grown in the western lagoon of Vonavona (locally known as Parara) Island in the Solomon Islands (8.221° S, 157.005° E). X-ray scans of the coral slab reveal clear annual growth rings, enabling the temporal distribution of skeletal material to be determined with an uncertainty of ± 1 year. The region experienced the April 2007 M_W 8.1 megathrust earthquake, which uplifted and killed the living coral surface, leaving approximately 60 cm of the colony exposed above sea level (Taylor et al., 2008). The collected slab measured $138 \times 100 \times 6$ cm, and its morphology records multiple exposed surfaces associated with emergence events during relative sea-level falls.

This study focuses on the 1997–2012 period, which includes three phases: (1) relative sea-level rise prior to the 2007 earthquake, (2) coseismic emergence during the earthquake, and (3) renewed relative sea-level rise in the postseismic period. Based on growth-band analysis, it is inferred that the microatoll subsided a minimum of 17 cm between 1997 and 2007, was uplifted approximately 63 cm during the earthquake, and subsequently subsided again by several centimeters. However, part or all of the observed sea-level change could also reflect absolute sea-level variations driven by global processes such as ocean circulation, glacial isostatic adjustment (GIA), and temperature-induced steric effect. In the western Solomon Islands, GIA and steric contributions to sea-level rise have been estimated at 7–10 mm/yr since 1993 (Albert et al., 2016). In addition, El Niño–Southern Oscillation (ENSO) events can produce sea-level falls of several tens of centimeters over one to two years. Consequently, using coral morphology alone to reconstruct relative sea-level changes can lead to erroneous interpretations: microatolls may not always emerge during sea-level drops, and growth may not keep pace with rapid sea-level rise.

To resolve these limitations, Gagan and colleagues (Gagan et al., 2008, 2009) introduced an innovative method based on the geochemical analysis of coral growth bands. This approach uses skeletal $\delta^{13}\text{C}$ as a proxy for vertical tectonic displacement. When shallow-living corals are suddenly uplifted, skeletal $\delta^{13}\text{C}$ shifts to more positive values. Conversely, subsidence leads to more negative values. The tectonic signal is large, exceeding the 2σ analytical uncertainty ($\sim 0.01\text{‰}$) by approximately 50–100 times. While shallow corals also display significant seasonal variability in $\delta^{13}\text{C}$ (driven by differences between daytime and nighttime lowest tides in summer vs. winter; (Buskirk et al., 1981) this variability is largely removed by annual averaging. The resulting inter-annual variation is small compared to tectonic signals.

Figure 5b illustrates this principle with $\delta^{13}\text{C}$ measurements spanning the period before, during, and after the April 2007 earthquake. The coseismic shift of 2‰ corresponds to an uplift of 73 cm (Taylor et al., 2008), yielding a transfer function of 36.5 cm/‰ . This estimate is derived by combining the vertical extent of the 2007 kill event with the measured water depth above the coral top at the time of collection. The minimum uplift, excluding the water depth component, is 63 cm (Figure 5).

Previous studies have also demonstrated a linear relationship between coral skeletal $\delta^{13}\text{C}$ and water depth, with decreasing $\delta^{13}\text{C}$ values at greater depth. This pattern has been reported in corals from diverse tropical settings and is attributed to reduced light penetration, lower photosynthetic activity, and declining irradiance with depth (Weber & Woodhead, 1970; Weber et al., 1976; Swart, 1983). Using the coral slice shown in Figure 5a, this relationship was tested by sampling the coral vertically.

High-resolution sampling was carried out with a computer-controlled drill along seven vertical transects (A–G), each perpendicular to the growth rings and extending below the 2007 exposed surface. A total of 60 samples were obtained, each representing a drilled volume of $1.5 \times 1.5 \times 1.0 \text{ mm}$. Samples were analyzed for both $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$. The oxygen isotope data provide an independent check on the influence of climate variability (e.g., ENSO events). Since the depths of samples were measured after the 2007 earthquake, they were corrected for postseismic offset relative to HLS. Although this correction alters the slope of the $\delta^{13}\text{C}$ –depth relationship, the correlation remains linear.

The results (Figure 5b) show a moderate but significant linear trend: skeletal $\delta^{13}\text{C}$ values decrease with increasing water depth. The correlation coefficient is $R^2 = 0.443$, confirming that depth exerts a measurable influence. The root-mean-square error (RMSE) indicates additional variability likely associated with sub-annual to annual sea-level changes. Nonetheless, the mean slope of 37.79 cm/‰ closely matches the coseismic estimate of 36.5 cm/‰ , providing confidence that $\delta^{13}\text{C}$ is a reliable proxy for relative sea-level reconstruction.

To further evaluate this approach, a horizontal transect (H) spanning the years 1997–2012 (Fig. 5a) was sampled and analyzed. This profile yielded 120 samples for combined $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ analysis. The long record enables reconstruction of pre-, co-, and postseismic vertical motion of the microatoll using the transfer function derived from vertical sampling. Because skeletal $\delta^{13}\text{C}$ is influenced by both climate and tectonics, the high-resolution coral dataset can be directly compared with independent sea-level time series. For this purpose, tide-gauge records from Honiara (9.429° S , 159.955° E ; Fig. 2b) were employed. The coral and tide-gauge sites are geographically close enough to record similar sea-level trends, including annual cycles and long-term rise, but far enough apart that the tide gauge remains unaffected by the same tectonic displacements. To enable direct comparison, the tide-gauge series was down-sampled and resampled to a common monthly grid consistent with the coral chronology.

Luc what have you done?

$$y = 37.686 (x) + -45.662$$

raw data
120 ???

4 Results and Discussion

4.1 Coseismic Deformation Determined from InSAR

The 2007 Solomon Islands earthquake deformation map derived from the ALOS PALSAR coseismic interferograms is shown in Figure 6a on a circular color scale. Here each fringe cycle (pink to pink, yellow to yellow, or green to green) indicates 20 cm of LOS deformation. In the motion-compensation imaging radar processor used in this study, the LOS direction is defined as from the satellite towards the ground. Therefore, uplift and westward motion lead to negative LOS values in ALOS PALSAR ascending geometry, whereas subsidence and eastward deformation lead to positive LOS values in the same geometry. Spatially continuous fringe patterns can be observed across multiple islands. This suggests that the proposed phase-filling strategy was able to successfully unwrap InSAR phase measurements over land surface pixels isolated by ocean pixels. The observed fringes over Vonavona and Kohingo are the densest, as these two islands are closest to the epicenter (Figure 2). A reduced deformation rate can be seen over New Georgia and land regions farther away from the epicenter. The same coseismic LOS deformation map is also presented on a continuous color map (Figure 6b), which highlights a maximum magnitude of -235.8 centimeters of LOS deformation measured on the southwestern coast of the Vonavona island, approximately 26 kilometers from the epicenter. Kohingo, another western island near the epicenter, experienced a maximum and minimum LOS deformation of -100.1 and -194.4 cm, respectively. Kolombangara, slightly northward of Vonavona, recorded LOS deformation ranging from -48.2 to -128.6 cm, and Rendova, slightly southward, recorded LOS deformation ranging from 11.8 to -69.5 cm. The deformation gradient is clear on the large New Georgia island, with the northwestern portion of the island near Kohingo experiencing the largest magnitude LOS deformation of -132.4 centimeters, while in the southeastern region a switch to positive LOS occurs near the most southeastern point of the island, registering a maximum positive LOS of 18.6 centimeters. Vangunu, the island farthest east in the study region, measured the largest positive LOS deformation of 25.6 centimeters. This value is of significantly smaller magnitude than that of Vonavona, as expected because of the loss of energy in the seismic waves as they travel farther away from the epicenter. Given the substantial horizontal and vertical deformation contributions and the lack of multiple LOS imaging geometry, it is not possible to further decompose the LOS deformation into horizontal and vertical components with a single LOS imaging geometry.

Three GPS stations are located within the coseismic map in Figure 6: MUND, RAVA, and HUSU. RAVA was used as the reference point to calibrate InSAR coseismic map, and the misfits between the InSAR LOS and the MUND and HUSU GPS stations were 3.2 and -3.9 centimeters, respectively (Table 4). The misfit values are within the expected tropospheric noise range of 2 to 5 centimeters of a single interferogram over a humid tropical islands (J. Chen et al., 2014). Additionally, the coseismic interferograms generated from acquisitions acquired from adjacent paths on different dates were joined without any visual discontinuities. This further confirms that the signal-to-noise ratio of the ALOS PALSAR coseismic deformation map is high. Previous research reported up to 2.46 ± 0.14 m of coastal coseismic uplift at Ranongga island based on surveys of coral reefs (Taylor et al., 2008). By comparison, the maximum LOS deformation at the nearby Vonavona island was determined to be 236 centimeters. The order of the in situ vertical measurements and the LOS measurements is consistent. As mentioned above, the lack of additional data does not enable a further decomposition of the LOS value to a vertical measurement for further comparison.

4.2 Postseismic Deformation Determined from InSAR

Figure 7 shows the average LOS rate (in cm/year) over three consecutive time periods of interest: (a) the first six months following the earthquake from April through

October 2007, (b) the following seven-month period covering October 2007 through April 2008, and (c) the remaining period for which SAR acquisitions are available that spans April 2008 through January 2010. In the green dashed box, the maximum LOS rate is observed near the epicenter over Vonavona with a consistent spatial pattern for all three time periods. At a given location, the observed LOS rate decreases over time. This is consistent with the temporal characteristics of the postseismic deformation signals. In contrast, the LOS patterns over the island regions outside the green dashed box are correlated in space but random in time – characteristics that align with tropospheric noise behavior as discussed in Section 3.3.

The average LOS rate was also calculated over the following growing time windows starting on April 16, 2007 through: (1) September 1, 2007; (2) October 17, 2007; (3) January 17, 2008; (4) March 3, 2008; (5) April 18, 2008; (6) September 3, 2008; (7) January 19, 2009; and (8) January 22, 2010 (Figure 8). The first stack (Figure 8(a)) included only a single interferogram (2007/04/16-2007/09/01), which shows the most prominent post-seismic deformation pattern over Vonavona. Inside the green dashed box, the observed LOS rate decreases over time (Figures 8(b)-(h)), and the spatial patterns are consistent in all eight stacks. Based on Equation (9) and Equation (10), both the postseismic deformation signal and tropospheric noise on the first SAR acquisition (04/16/2007) are expected to produce a spatially consistent pattern with reduced magnitude over time for the stacking formulation with growing time windows. Given that Figure 7(b) - (c) also display this consistent LOS pattern inside the green dashed box, and these two stacks do not include interferograms formed using the first SAR acquisition, it is concluded that the observed LOS pattern inside the green dashed box is likely associated with postseismic deformation signals. On the other hand, the observed LOS pattern over the other islands –Kolombangara, Kohinggo, and New Georgia—are correlated in space but random in time. This is consistent with the finding of Figure 7 that there are no detectable signals above tropospheric noise level here. As shown in Section 3.3, the residual tropospheric noise decreases as the total time span of an interferogram stack increases (Figures 8(a)-(h)).

Furthermore, the average LOS rate over the eight time windows defined in Figure 8 on Vonavona near the epicenter (Figure 9) are calculated. The postseismic deformation magnitude decay is greatest in the first 1.5 years after the earthquake, and then the deformation rate remains stable over the longer periods of time. The deformation rates from April 16, 2007 through (1) September 1, 2007, (2) October 17, 2007, (3) January 17, 2008, (4) March 3, 2008, (5) April 18, 2008, (6) September 3, 2008 are (1) 26.4 cm/year, (2) 19.5 cm/year, (3) 14.2 cm/year, (4) 12.7 cm/year, (5) 10.5 cm/year, and (6) 6.2 cm/year, respectively. The longer periods, spanning April 16, 2007 through January 19, 2009 and January 22, 2010 remain at 5.9 cm/year for both time periods, maintaining a similar magnitude as the previous window through September 3, 2008. This follows the expected exponential decay behavior associated with postseismic deformation rates. Due to the limited ALOS PALSAR acquisitions, how the surface deformed within the first several months after the earthquake can not be captured. The first and second postseismic ALOS acquisitions were on April 16, 2007 and September 1, 2007, with a ~ 5 months data gap. Nonetheless, the ability to capture the temporal evolution of the postseismic deformation with the sparse InSAR data set highlights the effectiveness of stacking for both time series generation and uncertainty quantification.

4.3 Coral $\delta^{13}C$ Time Series and Joint Interpretation

The unfiltered relative sea level changes time series derived from the $\delta^{13}C$ measurements of coral samples and the tide gauge (Figure 2(b)) are shown in Figure 10. The two data sets both capture similar seasonal weather signals and climate events (e.g. El Nino). However, the $\delta^{13}C$ -derived relative sea level changes also contains contributions from tectonic motion associated with the 2007 Solomon Island earthquake. The 2007 earthquake

is not expected to influence the tide gauge measurements recorded at Honiara, because it is too far away from the epicenter.

To remove seasonal weather-related signals, a moving average of 12 months was applied to the $\delta^{13}C$ time series (resampled on the monthly grid). The moving average filter was applied again to further remove signals related to changes in climate patterns (e.g. El Nino to La Nina) (Figure 11(a)). A clear vertical uplift is present during the time of the 2007 Solomon Islands earthquake. The aforementioned filtering was not applied to this coseismic section of the time series, which preserves the transient coseismic uplift signal. A vertical measurement of the visible die-down of a nearby coral sampling site (8.227°S, 157.012°E) from a previous study (Taylor et al., 2008) recorded uplift of 50 ± 15 centimeters. The coseismic uplift indicated from the $\delta^{13}C$ measurements of the study sample (8.221°S, 157.005°E) is approximately 66 centimeters, highlighting agreement between the two in situ measurement methods.

An L1 linear regression is applied to the time series prior to the earthquake to determine the interseismic subsidence rate. The slope of the interseismic subsidence is approximately 4 centimeters per year. This interseismic subsidence trend is not present in the tide gauge data, further highlighting that this signal was the result of tectonic motion rather than climate trends. Following the coseismic uplift, exponential decay of the vertical deformation is captured and emphasized by the polynomial-fitted trend line. The InSAR-derived post-seismic deformation rates from Figure 9 are provided with the post-seismic deformation captured by the coral for comparison (Figure 11(b)). The recorded dates for the InSAR deformation data correspond to the date of acquisition of the final SAR scene used in each growing time window discussed in Section 3.3–(1) September 1, 2007, (2) October 17, 2007, (3) January 17, 2008, (4) March 3, 2008, (5) April 18, 2008, (6) September 3, 2008, (7) January 19, 2009 and (8) January 22, 2010. While the $\delta^{13}C$ -derived measurements are plotted on a different axis due to them being purely vertical, the postseismic decay over time is in agreement with that captured by the post-seismic InSAR rates over Vonavona. These two data sets provide complementary information and together produce a comprehensive analysis of all three portions of the earthquake cycle.

5 Conclusion

This joint analysis provides information on the full earthquake cycle—inter-, co- and postseismic periods—surrounding the 2007 Mw 8.1 Solomon Islands earthquake. Phase filling allowed multiple islands from two separate ALOS PALSAR paths to be successfully unwrapped as a single interferogram without discontinuities or unwrapping errors. The coseismic InSAR map was calibrated and validated using three campaign GPS stations, and the resulting LOS deformation is of the same order of magnitude as previously published results. An advanced method proposed to detect postseismic deformation rates from signal with a small SNR due to tropospheric noise contamination successfully identified portions of the study region that contained a true deformation signal when there was a lack of in situ data for additional validation. A time series of postseismic deformation rates showed exponential decay of surface deformation following the earthquake, which follows the typical behavior often associated with postseismic processes. $\delta^{13}C$ measurements from a coral sample near Vonavona provided a record of the relative sea level in the region, which contained a signal from both tidal trends and vertical tectonic motion. Signal filtering techniques were employed to remove annual tidal signals and to smooth the data, producing a filtered signal that highlighted the vertical motion associated with tectonic activity. This signal captured interseismic subsidence, a sizable coseismic uplift, and a decay of postseismic deformation following the earthquake. These $\delta^{13}C$ results were compared with a control data set from a nearby tide gauge to highlight that the $\delta^{13}C$ signal contains tectonic contributions in addition to tidal trends.

This analysis was completed with a limited remote sensing data set—only 10 usable SAR acquisitions. The success in capturing postseismic deformation in a densely vegetated and humid region highlights the potential for future use of the proposed advanced method to detect signal for larger data sets, including those from the upcoming NiSAR mission. The combination of an L-band sensor for penetrating dense vegetation and the shorter revisit time of 12 days (rather than 46 days from ALOS PALSAR) will produce data sets that are more ideal for generating time series in similar regions. Furthermore, the results of coral $\delta^{13}C$ signal processing show promise as a method to obtain information on entire earthquake cycles, a time period that is rarely captured by a single data set. Identifying and leveraging complementary data sets, as was done in this joint analysis, enables a comprehensive analysis when both in situ and remote sensing data are limited.

Open Research Section

ALOS PALSAR SAR imagery over the New Georgia island chain in the Solomon Islands (Paths 343 and 344, Frames 7010 and 7020) can be downloaded from the Alaska Satellite Facility at <https://search.asf.alaska.edu>. All interferograms in this study were processed using a motion-compensation imaging radar processor (Zebker et al., 2010). Comparable interferograms can be generated with the following open-source InSAR packages: (1) InSAR Scientific Computing Environment 3 (ISCE3) (Rosen et al., 2018) or (2) GMTSAR (Sandwell et al., 2011).

Conflict of Interest

The authors declare there are no conflicts of interest for this manuscript.

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Table 1. ALOS PALSAR Acquisitions over the study site

Path	Frame	Data Available	Acquisitions
343	7010	December 28, 2006 - January 5, 2010	11
343	7020	December 28, 2006 - January 5, 2010	11
344	7010	January 14, 2007 - January 22, 2010	12
344	7020	January 14, 2007 - January 22, 2010	12

Table 2. Interferograms used for estimating the average postseismic deformation rate over three consecutive periods of interest

S_1	S_2	S_3
April 2007 - October 2007	October 2007 - April 2008	April 2008 - January 2010
04/16/2007-09/01/2007	10/17/2007-01/17/2008	04/18/2008-09/03/2008
04/16/2007-10/17/2007	10/17/2007-03/03/2008	04/18/2008-01/19/2009
09/01/2007-10/17/2007	10/17/2007-04/18/2008	04/18/2008-01/22/2010
	01/17/2008-03/03/2008	09/03/2008-01/19/2009
	01/17/2008-04/18/2008	09/03/2008-01/22/2010
	03/03/2008-04/18/2008	01/19/2009-01/22/2010

Table 3. Interferograms used for estimating the average postseismic deformation rate over three periods of interest without sharing any common SAR acquisitions

$S_{1'}$	$S_{2'}$	$S_{3'}$
April 2007 - October 2007	January 2008 - April 2008	September 2008 - January 2010
04/16/2007-09/01/2007	01/17/2008-03/03/2008	09/03/2008-01/19/2009
04/16/2007-10/17/2007	01/17/2008-04/18/2008	09/03/2008-01/22/2010
09/01/2007-10/17/2007	03/03/2008-04/18/2008	01/19/2009-01/22/2010

Table 4. GPS Validation for InSAR-Derived coseismic Deformation Map

Station	Location	InSAR LOS (cm)	GPS LOS (cm)	Misfit (cm)	Distance from RAVA (km)
RAVA	8.72°S, 157.41°E	7.17	7.17	0	0
MUND	8.33°S, 157.27°E	-86	-89.17	3.17	46.1
HUSU	8.50°S, 157.36°E	-37.54	-33.68	3.86	25.1

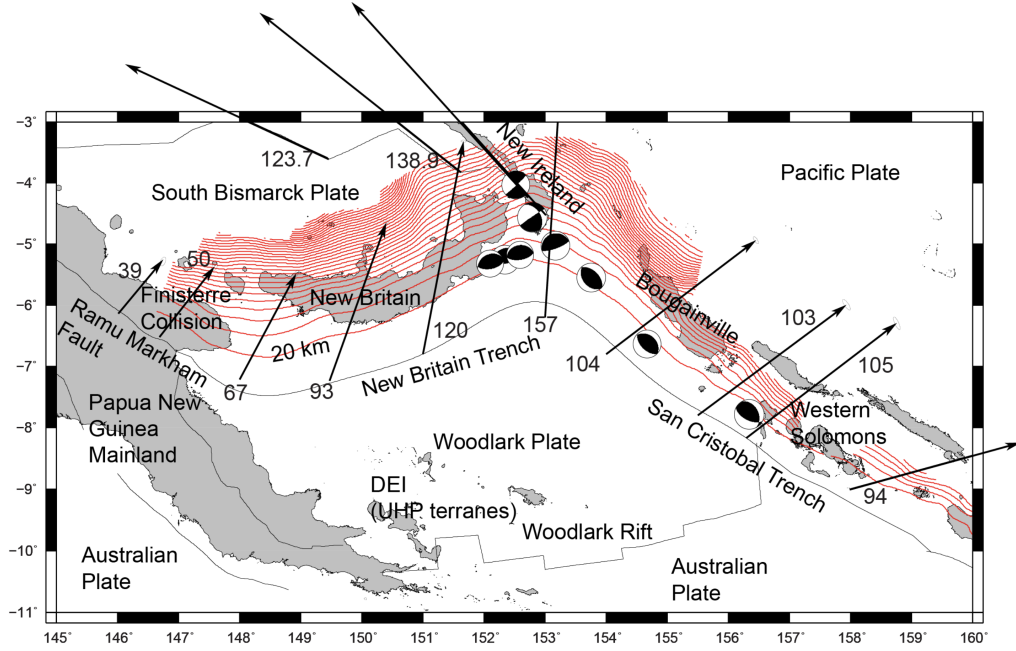


Figure 1. Convergence rates (black arrows, in mm/yr) and slab configuration (red contours) at the Ramu-Markham Fault, New Britain Trench, and San Cristobal Trench. Strike slip rates on the northern boundary of the South Bismarck Plate (e.g. on the Weitin fault that transects New Ireland) are also shown. Beach balls show focal mechanisms for earthquakes larger than Mw 7.5, 1999-present (from Global CMT catalog). New Britain slab contours are in 20 km intervals, with the maximum depth shown at 600 km (contours from (Hayes et al., 2012)). West of 153°E, New Britain Trench rates are determined using Woodlark/South Bismarck Plate relative motion (Wallace et al., 2004)), while east of 153°E rates are from Woodlark/Pacific Plate relative motion from (Wallace et al., 2014). DEI stands for D’Entrecasteaux Island, and UHP stands for Ultra-High Pressure.

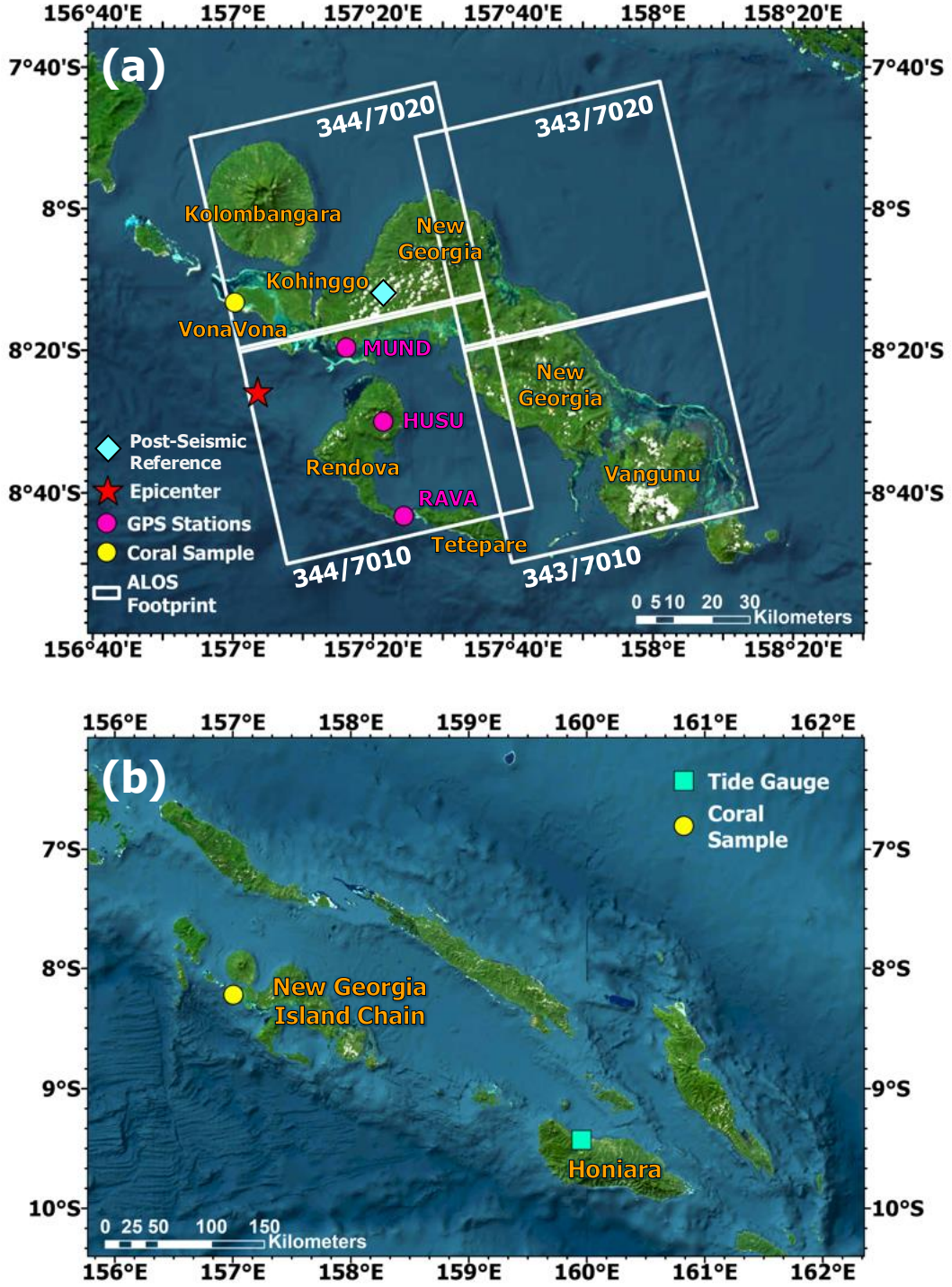


Figure 2. (a) Map of the study area over the Solomon Islands, specifically the New Georgia Island Chain. ALOS PALSAR footprints are outlined in white, and the path/frame numbers are labeled. Locations of three GPS stations, the coral sample site, and the epicenter of the April 1, 2007 Solomon Islands earthquake are labeled according to the legend. (b) The location of the coral sample site near Vonavona in the New Georgia Island chain and the location of the tide gauge off the coast of Honiara.

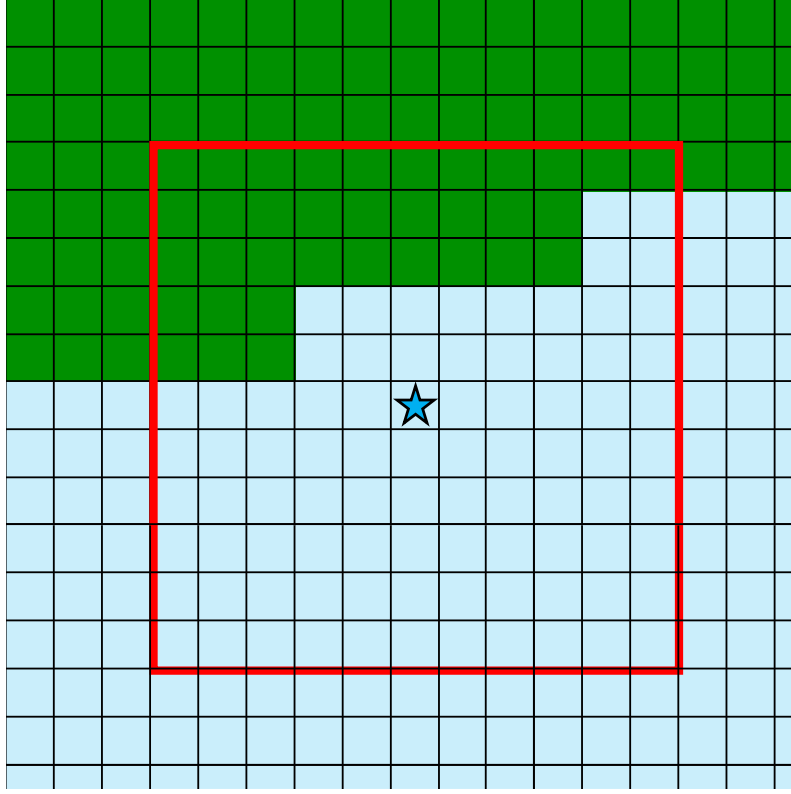


Figure 3. An illustration of phase filling at a water pixel near the shoreline. Blue pixels represent ocean pixels, and green pixels represent land pixels. To reduce large phase jumps at the ocean and land boundary and facilitate phase unwrapping, the wrapped phase value at an ocean pixel of interest (marked as the blue star) is replaced by the weighted average of wrapped phase observations at nearby land pixels within a square box of length 11 pixels centered at the ocean pixel (marked in red).

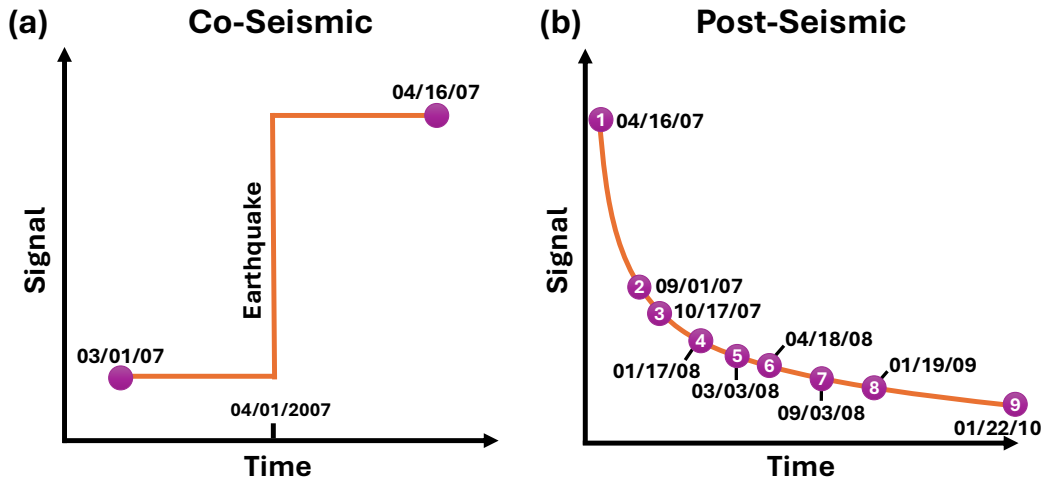


Figure 4. (a) The coseismic signal is modeled as a step function. This transient deformation can be captured by two SAR acquisitions that span the earthquake. (b) The postseismic deformation is expected to follow an exponential or logarithmic decay curve. High-quality Path 344 ALOS PALSAR acquisitions used in the postseismic study are marked as purple dots. Note that data gaps are present and the time interval between two nearest acquisitions varies.

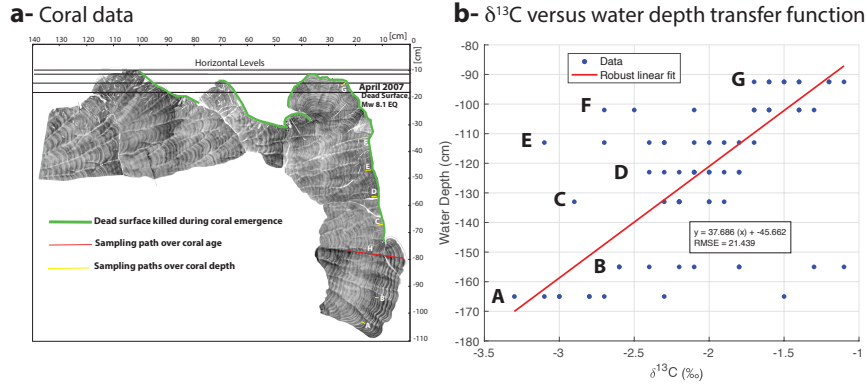


Figure 5. (a) X-ray radiograph of a ~ 1 cm-thick slice of fossil *Porites* coral collected in 2012, five years after the April 2007 M_W 8.1 earthquake. The green line marks the uplifted living surface, which was raised 63 cm above sea level. Reference nails were used to level the slab horizontally. Growth bands reveal vertical extension rates of 1–1.5 cm/yr. Earlier die-down surfaces, likely associated with emergence events, are visible in the 1950s–1970s. Vertical sampling transects (A–G, yellow) were drilled at 1.5 mm spacing (~ 6 samples/yr) for $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ analyses. Horizontal transect H captures 23 years of growth history across the 2007 seismic cycle. Ages of samples were determined by annual band counting from the living surface. (b) Transfer function between skeletal $\delta^{13}\text{C}$ and water depth derived from vertical sampling. This function is used to quantify tectonic uplift and subsidence associated with the 2007 event and longer-term microatoll growth.

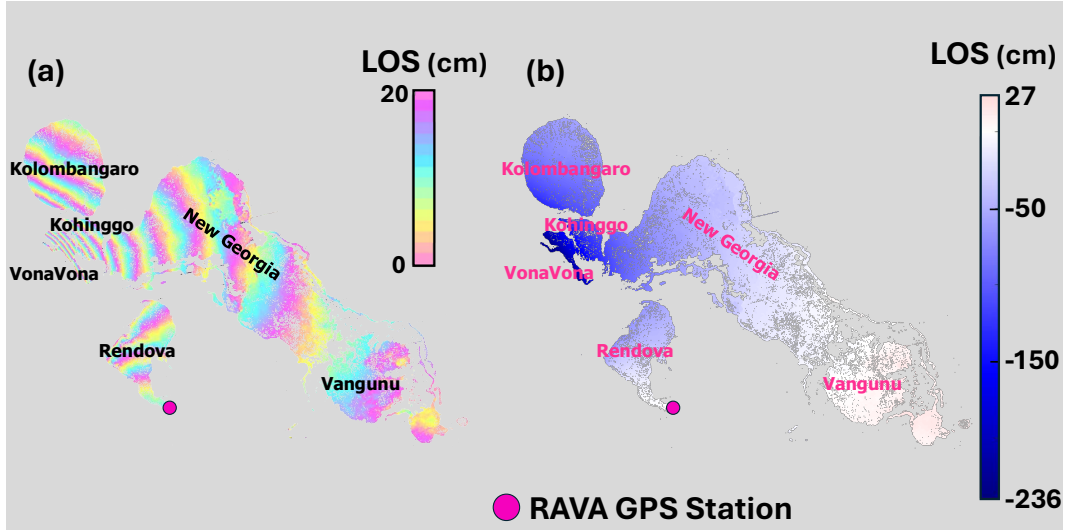


Figure 6. (a) Surface deformation associated with the 2007 Solomon Islands earthquake displayed on a circular color bar. Each fringe of corresponding colors, i.e. pink to pink, represents 20 centimeters of line-of-sight (LOS) deformation. Fringes appear denser near the epicenter. (b) The same InSAR LOS coseismic deformation map on a continuous color map. Blue (negative LOS values) indicates uplift and/or westward motion. The epicenter was identified southwest of Vonavona as shown in Figure 2.

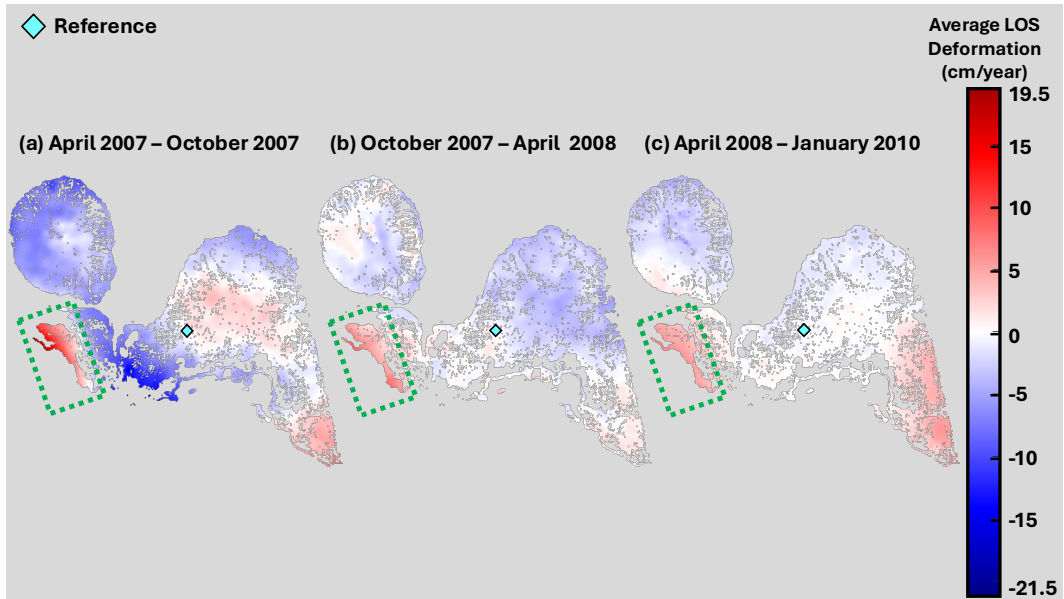


Figure 7. Average postseismic deformation rate estimates (cm/year) for the periods of (a) April 16, 2007 through October 17, 2007; (b) October 17, 2007 through April 18, 2008; and (c) April 18, 2008 through January 22, 2010. Subsidence and/or eastward motion lead to positive LOS motion (red). The LOS deformation rate is relative to the reference pixel located at 8.2°S , 157.4°E marked as a teal diamond. The results over Vonavona, highlighted within the green dashed box, reveal a consistent postseismic deformation pattern with decaying magnitude over time. Outside the green dashed box, the post seismic deformation signals are below the tropospheric noise level and thus undetectable using the ALOS PALSAR interferogram stacks.

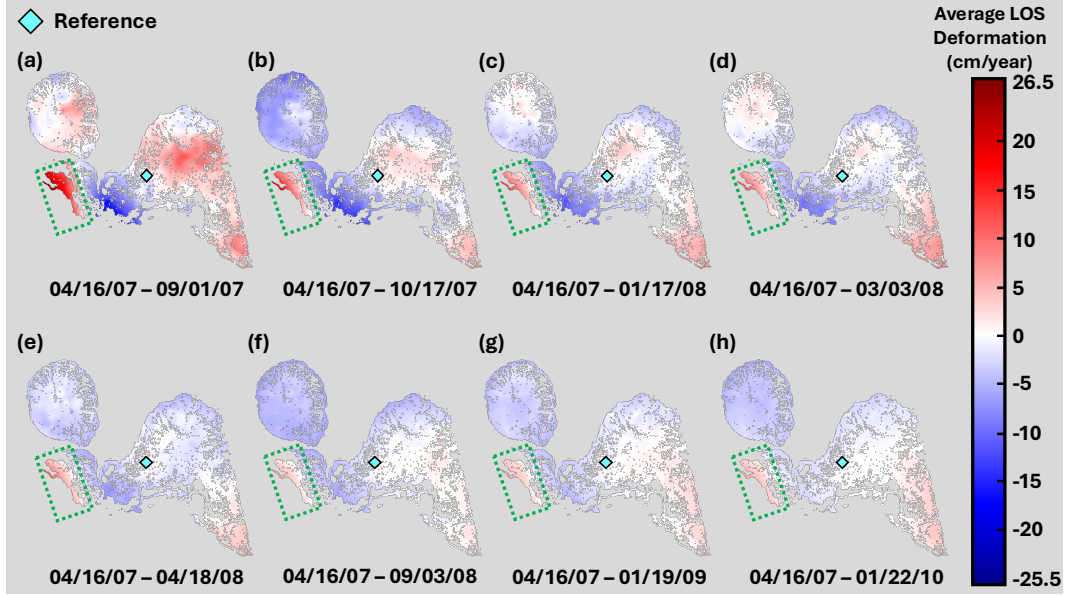


Figure 8. Average postseismic deformation rate estimates (in cm/year) for eight growing time windows starting on April 16, 2007 through: (a) September 1, 2007; (b) October 17, 2007; (c) January 17, 2008; (d) March 3, 2008; (e) April 18, 2008; (f) September 3, 2008; (g) January 19, 2009; and (h) January 22, 2010. Subsidence and/or eastward motion lead to positive LOS motion (red). The reference pixel located at 8.2°S, 157.4°E is marked by a teal diamond. Vonavona, outlined with the green dashed box, maintains a consistent postseismic deformation pattern with a magnitude that decays over time. Outside the green dashed box, the post seismic deformation signals are below the tropospheric noise level and thus undetectable using the ALOS PALSAR interferogram stacks.

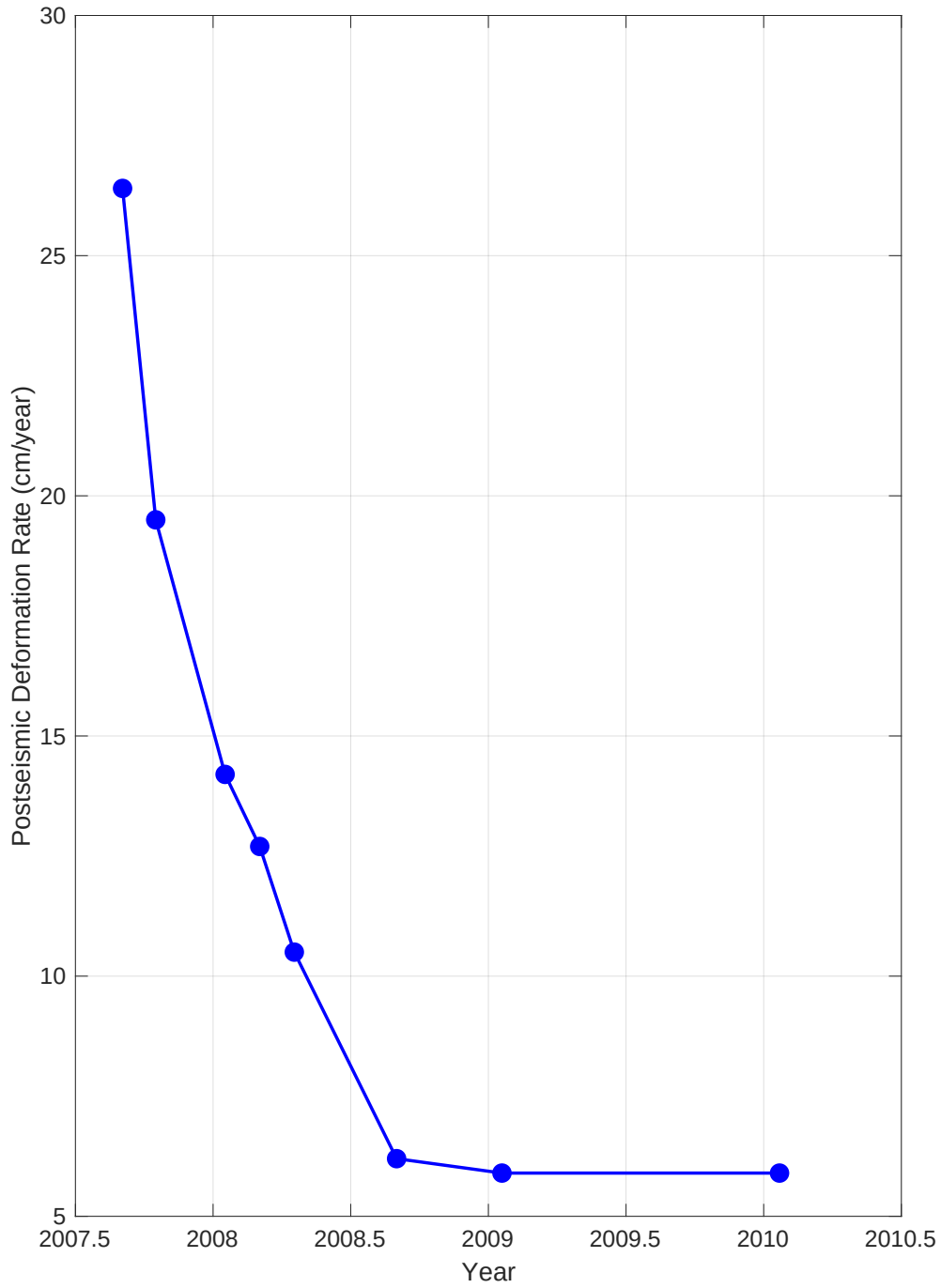


Figure 9. The average deformation rate (cm / year) of time windows spanning April 16, 2007 through: (1) September 1, 2007; (2) October 17, 2007; (3) January 17, 2008; (4) March 3, 2008; (5) April 18, 2008; (6) September 3, 2008; (7) January 19, 2009; and (8) January 22, 2010. The reported date on the x-axis corresponds to the date of acquisition of the final SAR scene used in each respective window. The plot appears similar to that of an exponential decay - the trend often used to describe postseismic deformation processes.

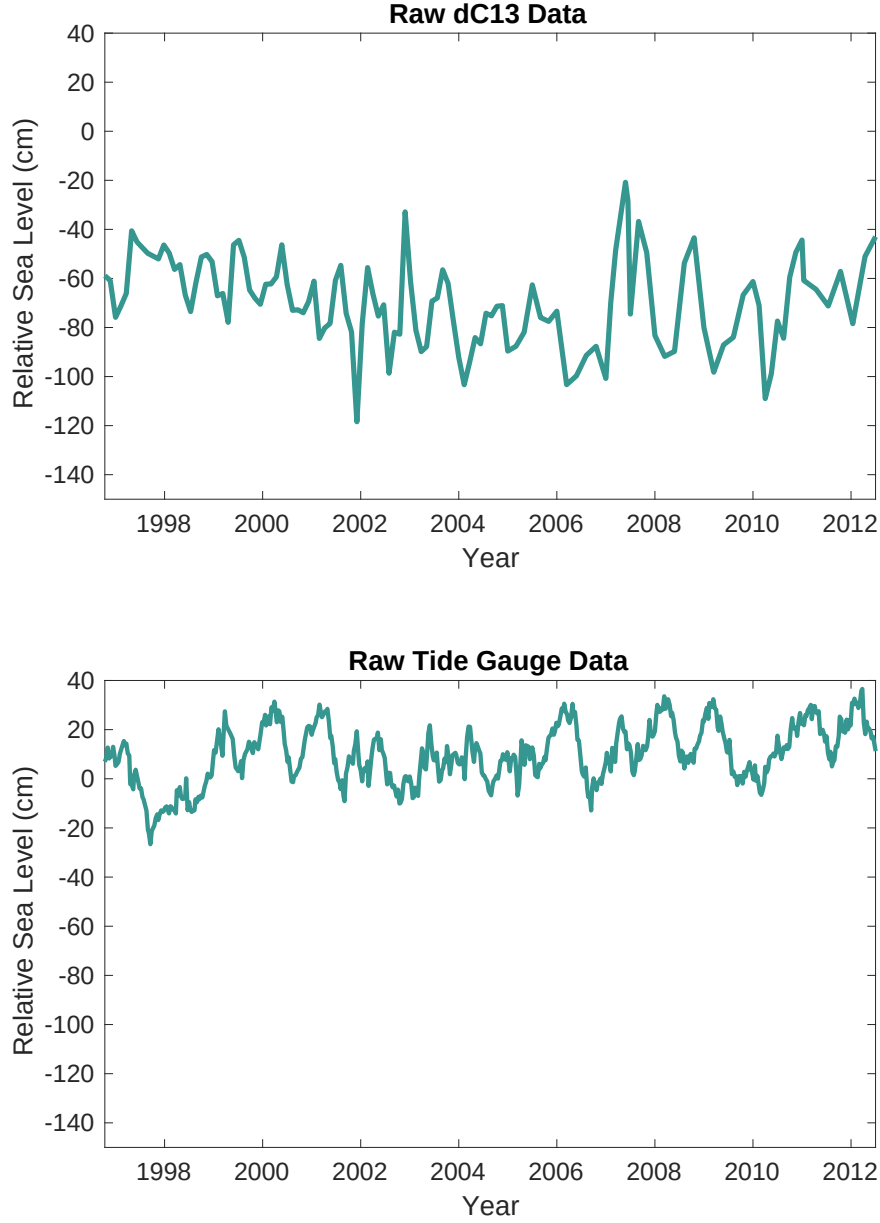


Figure 10. Changes in relative sea level (cm) over time captured by the coral ($\delta^{13}C$) data and tide gauge data. The coral sample was collected at -8.22° S, 157.01° E. The tide gauge data was collected at the Honiara-B station (-9.43° S, 159.96° E). Both data sets contain signals from climate events and seasonal weather variations, and the coral-derived data additionally contains vertical tectonic motion associated with the 2007 Solomon Islands earthquake.

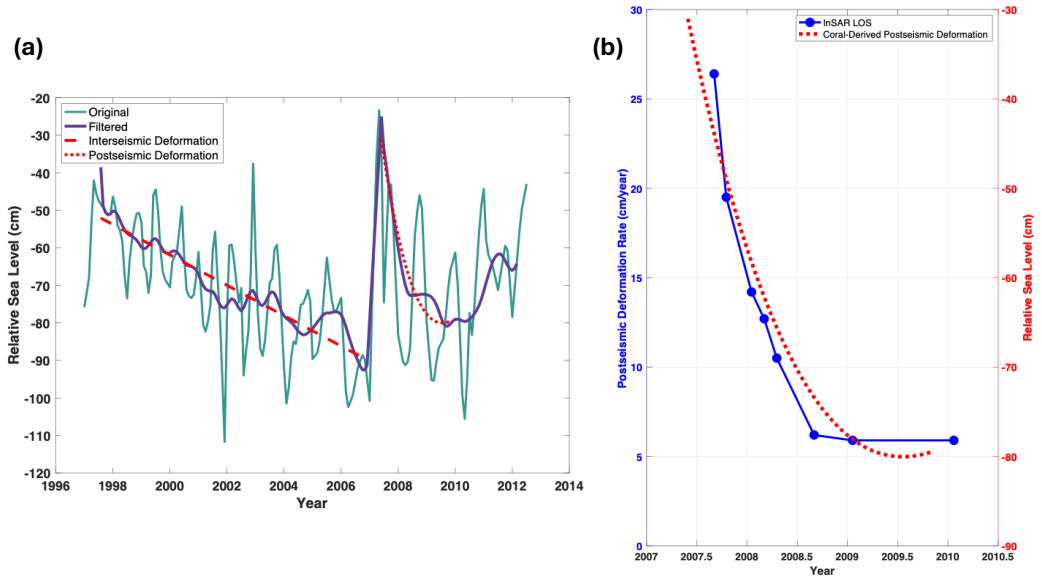


Figure 11. (a) The original (green) and filtered (purple) relative sea level change time series (in cm) as derived from Coral $\delta^{13}C$ measurements. The red dashed line shows a subsidence trend fitted pre-earthquake using an L1 linear regression. A large uplift is visible during the earthquake, which is followed by post-seismic subsidence, denoted by the polynomial-fitted dotted red line. (b) The blue line denotes the postseismic deformation rate captured by ALOS interferograms as displayed in Figure 9. The red dotted line represents the post-seismic deformation trend as fitted in (a). The dates for the InSAR-derived data correspond to the acquisition date of the final SAR scene utilized in each respective time window.